

2.6–2.7 LU, LDU, Permutations, and Transposes

Concise exam revision notes for elimination factorizations and matrix operation rules.

Based on handwritten MIT 18.06 revision notes.

1 Core Idea

Core idea. Elimination can be stored as a factorization. Instead of redoing row operations, write the matrix as products of simple matrices.

Without row exchanges, elimination gives

$$A = LU.$$

If the pivots are separated into a diagonal matrix D , then

$$A = LDU.$$

Here L stores elimination multipliers, D stores pivots, and U stores the final upper triangular structure.

2 Solving With LU

To solve

$$Ax = b, \quad A = LU,$$

do two triangular solves:

$$Lc = b, \quad Ux = c.$$

Step	Equation	Meaning
Forward substitution	$Lc = b$	Apply stored elimination.
Back substitution	$Ux = c$	Solve the upper triangular system.

Exam memory. LU is useful because solving triangular systems is faster than starting elimination again.

3 Permutations

If a row exchange is needed, use a permutation matrix:

$$PA = LU.$$

Permutation matrices reorder rows and satisfy

$$P^{-1} = P^T, \quad P^T P = I.$$

Fact	Meaning
$P^{-1} = P^T$	Undoing a permutation is another permutation.
$P^T P = I$	Permutations preserve lengths and dot products.
PA	Reorders rows of A .
AP	Reorders columns of A .

4 Transpose and Inverse Rules

Order reverses when transposing or inverting products:

$$(AB)^T = B^T A^T, \quad (AB)^{-1} = B^{-1} A^{-1}.$$

For an inverse transpose,

$$(A^{-1})^T = (A^T)^{-1}.$$

Common mistake. Do not transpose a product in the same order. The order must reverse.

5 Symmetric Matrices

A matrix is symmetric when

$$A^T = A.$$

The product $A^T A$ is always symmetric:

$$(A^T A)^T = A^T A.$$

The dot-product identity behind this is

$$(Ax)^T y = x^T A^T y.$$

6 Block Transposes

Transpose works block by block and also swaps block positions:

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix}^T = \begin{bmatrix} A^T & C^T \\ B^T & D^T \end{bmatrix}.$$

This is the same rule as ordinary transposes: rows become columns.

7 Banded Matrices

Banded matrices have nonzeros near the diagonal, so elimination can be cheaper when the band structure is preserved. A common example is tridiagonal structure:

$$\begin{bmatrix} * & * & 0 & 0 \\ * & * & * & 0 \\ 0 & * & * & * \\ 0 & 0 & * & * \end{bmatrix}.$$

The exam habit is to use the structure instead of treating every entry as dense.

8 Common Mistakes

Mistake	Correct exam habit
Forgetting row exchanges in LU.	Use $PA = LU$ when pivots require swaps.
Solving $LUx = b$ in one jump.	First solve $Lc = b$, then $Ux = c$.
Writing $(AB)^T = A^T B^T$.	Reverse order: $B^T A^T$.
Assuming $A^T A = AA^T$.	Both are symmetric, but usually different.
Ignoring zeros in a banded matrix.	Use the sparsity pattern to simplify work.

9 Final Exam Checklist

Checklist.

1. Check whether row exchanges are needed.
2. Use $A = LU$ if no row exchanges occur.
3. Use $PA = LU$ if row exchanges occur.
4. Solve $Lc = b$ by forward substitution.
5. Solve $Ux = c$ by back substitution.
6. Reverse order when transposing or inverting products.
7. Check symmetry with $A^T = A$.
8. Use banded or triangular structure when present.

Permutations, Subspaces, Rank, and Dot Products

Concise revision notes from the 30 Dec handwritten matrix-operations review.

Based on handwritten MIT 18.06 revision notes.

1 Core Idea

Core idea. Early linear algebra questions often reduce to three checks: what operations preserve structure, whether a set is a subspace, and how rank controls solution behavior.

This note collects matrix-operation facts used repeatedly later: permutation matrices, subspaces, rank cases, row/column relationships, dot products, and skew-symmetry.

2 Permutation Matrices

A permutation matrix reorders rows or columns. It is orthogonal:

$$P^T P = I, \quad P^{-1} = P^T.$$

Multiplying on the left reorders rows:

$$PA.$$

Multiplying on the right reorders columns:

$$AP.$$

Exam memory. A permutation matrix looks like the identity matrix with its rows or columns rearranged.

3 Subspace Test

A set S is a subspace when:

1. $0 \in S$;
2. if $u, v \in S$, then $u + v \in S$;
3. if $u \in S$, then $cu \in S$.

The nullspace is a subspace because if

$$Au = 0, \quad Av = 0,$$

then

$$A(u + v) = 0, \quad A(cu) = 0.$$

Common mistake. A set defined by $Ax = b$ with $b \neq 0$ is usually not a subspace, because it does not contain the zero vector.

4 Rank Cases

For $A \in \mathbb{R}^{m \times n}$, rank r controls solution behavior.

Rank case	Meaning	Exam consequence
$r = n$	Full column rank.	At most one solution to $Ax = b$.
$r = m$	Full row rank.	$Ax = b$ is solvable for every $b \in \mathbb{R}^m$.
$r = m = n$	Full rank square.	Exactly one solution for every b .
$r < m$ and $r < n$	Not full rank.	Zero or infinitely many solutions.

5 Row Space and Column Space

Row space and column space have the same dimension:

$$\dim C(A) = \dim C(A^T) = \text{rank}(A).$$

But they usually live in different ambient spaces:

$$C(A) \subseteq \mathbb{R}^m, \quad C(A^T) \subseteq \mathbb{R}^n.$$

Exam memory. Same dimension does not mean the same space. Check where the vectors live.

6 Dot Product

The dot product is

$$x \cdot y = x^T y.$$

It measures alignment:

$$x^T y = 0$$

means the vectors are perpendicular.

It also interacts cleanly with matrix multiplication:

$$(Px) \cdot (Py) = x \cdot y$$

for a permutation or orthogonal matrix P .

7 Skew-Symmetric Matrices

A matrix is skew-symmetric when

$$A^T = -A.$$

The diagonal entries must be zero, because

$$a_{ii} = -a_{ii} \Rightarrow a_{ii} = 0.$$

Off-diagonal entries come in opposite pairs.

8 Common Mistakes

Mistake	Correct exam habit
Treating $Ax = b$ as a subspace when $b \neq 0$.	Check whether the zero vector is included.
Confusing row operations with column operations.	Left multiply changes rows; right multiply changes columns.
Assuming equal dimensions imply equal spaces.	Check the ambient spaces.
Forgetting the full-rank square case.	$r = m = n$ gives existence and uniqueness.
Putting nonzero entries on a skew-symmetric diagonal.	The diagonal must be zero.

9 Final Exam Checklist

Checklist.

1. Identify whether multiplication is on the left or right.
2. Check $P^T P = I$ for permutation matrices.
3. Test subspaces by zero vector, addition, and scalar multiplication.
4. Write the size $m \times n$ before using rank facts.
5. Find rank r .
6. Use full column rank for uniqueness.
7. Use full row rank for existence.
8. Use $x^T y = 0$ to check perpendicularity.
9. Check skew-symmetry with $A^T = -A$.

3.2–3.3 Nullspace, Special Solutions, and Complete Solutions

Concise exam revision notes for homogeneous and nonhomogeneous systems.

Based on handwritten MIT 18.06 revision notes.

1 Core Idea

Core idea. Row reduction separates a linear system into forced variables and free variables. Pivot variables are forced by the equations; free variables create the directions in the nullspace.

The two central systems are:

$$Ax = 0$$

and

$$Ax = b.$$

The first finds input directions that disappear under A . The second asks whether the right side b can be built from the columns of A .

2 Big Picture

Let A be an $m \times n$ matrix.

Question	Meaning	Output
$Ax = 0$	Which input vectors are sent to zero?	The nullspace $N(A)$.
$Ax = b$	Can the columns of A produce this b ?	Zero, one, or infinitely many solutions.
Row reduction	Reveals pivots and free variables.	The structure of all solutions.

The rank r is the number of pivots. The number of free variables is

$$n - r.$$

Exam trigger. If a problem says “find all solutions”, expect an answer of the form

$$x = x_p + x_n.$$

Here x_p is one particular solution and x_n runs through the nullspace.

3 Nullspace vs Column Space

Space	Definition	Lives in	What it tells you
Nullspace $N(A)$	All x such that $Ax = 0$.	$\mathbb{R}^n$	Directions that do not change Ax .
Column space $C(A)$	All linear combinations of the columns of A .	$\mathbb{R}^m$	Right sides b for which $Ax = b$ is solvable.

Why the dimensions differ:

- x has one entry per column of A , so $x \in \mathbb{R}^n$.
- Ax and b have one entry per row of A , so they live in $\mathbb{R}^m$.
- Therefore $N(A) \subseteq \mathbb{R}^n$, while $C(A) \subseteq \mathbb{R}^m$.

Common mistake. Do not place the nullspace in $\mathbb{R}^m$. The nullspace contains input vectors, so its vectors have length n .

4 Free Variables and Special Solutions

Free variables appear in columns without pivots. Each free variable gives one independent nullspace direction.

4.1 Solving pattern

1. Row-reduce A to echelon or reduced echelon form.
2. Mark pivot variables and free variables.

3. Set one free variable to 1 and all other free variables to 0.
4. Solve for the pivot variables.
5. The resulting vector is one special solution.
6. Repeat once for each free variable.

If the free variables are x_3 and x_4 , use

$$(x_3, x_4) = (1, 0) \quad \text{and} \quad (x_3, x_4) = (0, 1).$$

These choices produce two special solutions s_1 and s_2 .

Core idea. The special solutions form a basis for the nullspace:

$$N(A) = \text{span}\{s_1, \dots, s_{n-r}\}.$$

Exam trigger. If $n > m$, then $r \leq m < n$. There must be at least one free variable, so $Ax = 0$ has a nonzero solution.

5 Complete Solution

For $Ax = b$, first check consistency. If the system is consistent, all solutions are

$$x = x_p + x_n.$$

Part	Equation	Role	How to find it
Particular solution x_p	$Ax_p = b$	One fixed solution.	Often set free variables to 0.
Nullspace part x_n	$Ax_n = 0$	Directions added to x_p .	Combine special solutions.

If $s_1, \dots, s_k$ are special solutions, then

$$x = x_p + c_1 s_1 + c_2 s_2 + \dots + c_k s_k.$$

The constants $c_1, \dots, c_k$ are arbitrary.

Why this works:

$$A(x_p + x_n) = Ax_p + Ax_n = b + 0 = b.$$

Adding a nullspace vector keeps the same right side.

Common mistake. Do not multiply the particular solution by an arbitrary constant. The free constants belong only to nullspace directions.

6 Rank, Pivots, and Free Variables

Quantity	Meaning	Exam use
$r = \text{rank}(A)$	Number of pivots.	Counts independent columns.
Pivot variables	Variables solved from equations.	Forced by the system.
Free variables	Variables chosen freely.	Create special solutions.
$n - r$	Number of free variables.	Dimension of $N(A)$.

The rank-nullity count used here is

$$\dim N(A) = n - r.$$

Exam memory. Full column rank controls uniqueness. Full row rank controls existence. Square full rank gives both.

For $Ax = 0$, row operations do not change the solution set, so it is enough to solve

$$Rx = 0.$$

For $Ax = b$, reduce the augmented matrix $[A \ b]$. A zero coefficient row with a nonzero augmented entry means no solution.

7 Full Column Rank vs Full Row Rank

Case	Condition	Nullspace	Behavior of $Ax = b$
Square full rank	$r = m = n$	Only 0.	Exactly one solution for every b .
Full column rank	$r = n \leq m$	Only 0.	Zero or one solution.
Full row rank	$r = m \leq n$	Dimension $n - m$.	At least one solution for every b .
Not full rank	$r < m$ and $r < n$	Dimension $n - r$.	Zero or infinitely many solutions.

7.1 Full column rank

Every column has a pivot.

- No free variables.
- $N(A) = \{0\}$.
- If a solution exists, it is unique.
- Some right sides b may still be outside the column space.

7.2 Full row rank

Every row has a pivot.

- The column space is all of $\mathbb{R}^m$.
- $Ax = b$ is solvable for every $b \in \mathbb{R}^m$.
- If $n > m$, free variables make the solution infinite.

Common mistake. Full column rank gives uniqueness, not automatic existence. Full row rank gives existence for every b , not uniqueness.

8 Rank-One Matrix Example

A rank-one matrix has the form

$$A = uv^T.$$

Then

$$Ax = u(v^T x).$$

If $u \neq 0$, then $Ax = 0$ is equivalent to

$$v^T x = 0.$$

So the nullspace consists of vectors perpendicular to v .

Example:

$$A = \begin{bmatrix} 1 & 3 & 10 \\ 2 & 6 & 20 \\ 3 & 9 & 30 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \begin{bmatrix} 1 & 3 & 10 \end{bmatrix}.$$

The nullspace condition is

$$x_1 + 3x_2 + 10x_3 = 0.$$

Free choice	Resulting special solution
$x_2 = 1, x_3 = 0$	$(-3, 1, 0)$
$x_2 = 0, x_3 = 1$	$(-10, 0, 1)$

Therefore

$$N(A) = \text{span}\{(-3, 1, 0), (-10, 0, 1)\}.$$

Common mistake. For $A = uv^T$, the nullspace condition is $v^T x = 0$, not $u^T x = 0$. The vector v matches the input dimension.

9 Common Mistakes

Mistake	Correct exam habit
Putting $N(A)$ in $\mathbb{R}^m$.	$N(A) \subseteq \mathbb{R}^n$, because x has n entries.
Counting rows to find special solutions.	Count free variables: $n - r$.
Giving only x_p for $Ax = b$.	Add the whole nullspace: $x = x_p + x_n$.
Scaling x_p by a free constant.	Only special solutions get arbitrary constants.
Treating full column rank as automatic solvability.	It gives uniqueness if a solution exists.
Treating full row rank as uniqueness.	It gives existence for every b ; free variables may remain.
Ignoring inconsistent rows.	Check $[A \ b]$ before writing a solution.

10 Final Exam Checklist

Checklist.

1. Write the size: A is $m \times n$.
2. Row-reduce and find the rank r .
3. Mark pivot variables and free variables.
4. For $Ax = 0$, create one special solution per free variable.
5. Check that the number of special solutions is $n - r$.
6. For $Ax = b$, reduce $[A \ b]$ and check consistency.
7. Find one particular solution x_p .
8. Add the nullspace:

$$x = x_p + c_1 s_1 + \cdots + c_k s_k.$$

9. Check dimensions: $x \in \mathbb{R}^n$, $b \in \mathbb{R}^m$.
10. State the outcome directly: no solution, one solution, or infinitely many.

Complete Solutions Worked Example

Concise revision notes for reading free variables, special solutions, and $x = x_p + x_n$.

Based on handwritten MIT 18.06 revision notes.

1 Core Idea

Core idea. A system with free variables has infinitely many solutions. One particular solution solves $Ax = b$; special solutions describe all movement inside $N(A)$.

The final answer has the form

$$x = x_p + x_n.$$

If $s_1, \dots, s_k$ are special solutions, then

$$x = x_p + c_1 s_1 + \dots + c_k s_k.$$

2 Reading Free Variables

Free variables come from columns without pivots. In a reduced system, set each free variable one at a time to create nullspace directions.

Variable type	Meaning
Pivot variable	Forced by the equations.
Free variable	Chosen freely; creates a nullspace direction.
Particular choice	Usually set free variables to simple values to solve $Ax = b$.

3 Particular Solution

For a consistent system

$$Ax = b,$$

choose convenient free variable values, often zeros, and solve for pivot variables. This gives one particular solution

$$x_p.$$

It is not the whole answer unless the nullspace is zero.

4 Special Solutions

To find the nullspace, solve

$$Ax = 0.$$

For each free variable:

1. set that free variable to 1;
2. set the other free variables to 0;
3. solve for pivot variables.

The resulting vectors $s_1, \dots, s_k$ form a basis for $N(A)$:

$$N(A) = \text{span}\{s_1, \dots, s_k\}.$$

5 Putting the Answer Together

If one particular solution is x_p and the nullspace basis is s_1, s_2 , write

$$x = x_p + c_1 s_1 + c_2 s_2.$$

The constants c_1, c_2 are arbitrary.

Meaning. The particular solution reaches b . The nullspace part moves without changing Ax .

6 Common Mistakes

Mistake	Correct exam habit
Giving only x_p .	Add the full nullspace.
Using $Ax = b$ to find special solutions.	Use $Ax = 0$.
Multiplying x_p by arbitrary constants.	Constants multiply only nullspace directions.
Forgetting one free variable.	Number of special solutions equals number of free variables.
Choosing complicated free values.	Use 0's and 1's unless the system suggests otherwise.

7 Final Exam Checklist

Checklist.

1. Row-reduce the augmented matrix $[A \ b]$.
2. Check consistency.
3. Mark pivot variables and free variables.
4. Choose simple free-variable values to get x_p .
5. Solve $Ax = 0$ for special solutions.
6. Create one special solution per free variable.
7. Write $x = x_p + c_1s_1 + \cdots + c_ks_k$.
8. Check by substituting: $Ax_p = b$ and $As_i = 0$.

3.4–3.5 Independence, Basis, Dimension, and Subspaces

Concise exam revision notes for independence, span, bases, and subspace dimensions.

Based on handwritten MIT 18.06 revision notes.

1 Core Idea

Core idea. Independence, span, and basis are all controlled by the equation $Ax = 0$, where the columns of A are the vectors being tested.

For vectors $v_1, \dots, v_n$, put them into a matrix

$$A = [v_1 \quad v_2 \quad \cdots \quad v_n].$$

Then the dependence test is

$$x_1v_1 + x_2v_2 + \cdots + x_nv_n = 0 \quad \Longleftrightarrow \quad Ax = 0.$$

The vectors are independent exactly when the only solution is $x = 0$.

2 Independence and Dependence

Situation	Meaning	Exam consequence
Independent columns	$Ax = 0$ has only $x = 0$.	No free variables.
Dependent columns	$Ax = 0$ has a nonzero solution.	At least one free variable.
More columns than rows	$n > m$.	Columns must be dependent.

Exam trigger. To test whether vectors are independent, solve $Ax = 0$, not $Ax = b$.

If $Ax = 0$ has a nonzero solution, that solution gives an explicit linear relation among the columns. For example,

$$x_1v_1 + x_2v_2 + x_3v_3 = 0$$

with some nonzero coefficient means one vector is forced by the others.

3 Span

The span is the set of all linear combinations of the given vectors:

$$\text{span}\{v_1, \dots, v_k\} = \{c_1v_1 + \cdots + c_kv_k\}.$$

It answers the question: which vectors can be produced?

Vectors in $\mathbb{R}^2$	Span
$(1, 0)^T, (0, 1)^T$	All of $\mathbb{R}^2$.
$(1, 0)^T, (-1, 0)^T$	Only the x -axis.

Common mistake. Having two vectors in $\mathbb{R}^2$ does not automatically mean they span $\mathbb{R}^2$. They must point in independent directions.

4 Basis

A basis is a set of vectors that is both independent and spanning.

Property	What it prevents	What it guarantees
Independent	No repeated direction.	No vector is unnecessary.
Spanning	No missing direction.	Every vector in the space is reachable.
Basis	Both at once.	Coordinates are unique.

Exam memory. A basis is the smallest spanning set and the largest independent set.

5 Row Space and Column Space

For an $m \times n$ matrix A , the column vectors have length m , while row vectors have length n .

Space	Built from	Lives in	Basis rule
Column space $C(A)$	Columns of A .	$\mathbb{R}^m$	Use pivot columns of the original A .
Row space $C(A^T)$	Rows of A .	$\mathbb{R}^n$	Use nonzero rows of echelon form R .

Example:

$$A = \begin{bmatrix} 2 & 4 \\ 3 & 6 \end{bmatrix}, \quad R = \begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix}.$$

The row space can be represented from R :

$$C(A^T) = \text{span}\{(1, 2)\}.$$

The column space must use the pivot column from the original matrix:

$$C(A) = \text{span} \left\{ \begin{bmatrix} 2 \\ 3 \end{bmatrix} \right\}.$$

Common mistake. Do not use the pivot columns of R as a basis for $C(A)$. Row operations change the columns, but they preserve the row space.

6 Dimension and Rank

The dimension of a subspace is the number of vectors in a basis for that subspace.

Quantity	Value	Meaning
$\dim C(A)$	r	Number of pivot columns.
$\dim C(A^T)$	r	Number of nonzero rows in R .
$\dim N(A)$	$n - r$	Number of free variables.
$\dim N(A^T)$	$m - r$	Number of missing pivots among rows.

The row space and column space have the same dimension:

$$\dim C(A) = \dim C(A^T) = r.$$

They usually live in different spaces, so they are not the same subspace.

7 Four Fundamental Subspaces

For $A \in \mathbb{R}^{m \times n}$, the four subspaces are:

Subspace	Lives in	Dimension	How to find a basis
Column space $C(A)$	$\mathbb{R}^m$	r	Pivot columns of A .
Row space $C(A^T)$	$\mathbb{R}^n$	r	Nonzero rows of R .
Nullspace $N(A)$	$\mathbb{R}^n$	$n - r$	Special solutions of $Ax = 0$.
Left nullspace $N(A^T)$	$\mathbb{R}^m$	$m - r$	Special solutions of $A^T y = 0$.

Dimension check. The dimensions split by ambient space: $r + (n - r) = n$ in $\mathbb{R}^n$, and $r + (m - r) = m$ in $\mathbb{R}^m$.

8 Rank-One Reminder

A rank-one matrix is one column times one row:

$$A = uv^T.$$

For example,

$$\begin{bmatrix} a \\ b \\ c \end{bmatrix} \begin{bmatrix} 2 & 3 & 7 & 8 \end{bmatrix}$$

has rank 1 if both factors are nonzero.

Exam use. Rank-one matrices are useful because all columns are multiples of one vector and all rows are multiples of one row.

9 Common Mistakes

Mistake	Correct exam habit
Testing independence with $Ax = b$.	Test with $Ax = 0$.
Forgetting that $n > m$ forces dependence.	More vectors than dimension means at least one free variable.
Using columns of R for $C(A)$.	Use pivot columns of original A .
Saying row space and column space are the same.	They have the same dimension r , but may live in different spaces.
Counting all columns as a basis.	Count pivot columns only.

10 Final Exam Checklist

Checklist.

1. Write the size: A is $m \times n$.
2. Put the vectors as columns if testing independence or span.
3. Solve $Ax = 0$ to test independence.
4. Find the rank r from pivots.
5. Use pivot columns of A for a basis of $C(A)$.
6. Use nonzero rows of R for a basis of $C(A^T)$.
7. Use special solutions for a basis of $N(A)$.
8. Check dimensions: r , r , $n - r$, and $m - r$.
9. State whether the vectors are independent, spanning, a basis, or dependent.

4.1 Orthogonality and the Four Fundamental Subspaces

Concise exam revision notes for orthogonal complements in $\mathbb{R}^n$ and $\mathbb{R}^m$.

Based on handwritten MIT 18.06 revision notes.

1 Core Idea

Core idea. Orthogonality connects the four fundamental subspaces in pairs. Row space is perpendicular to nullspace; column space is perpendicular to left nullspace.

The dot product test is

$$v^T w = 0.$$

If this holds for every vector v in one subspace and every vector w in another, the two subspaces are orthogonal.

2 Orthogonal Subspaces vs Orthogonal Complements

Orthogonal subspaces may intersect only at zero, but that alone does not mean they fill the whole space.

Idea	Meaning	Exam check
Orthogonal subspaces	Every vector in one is perpendicular to every vector in the other.	Dot products are zero.
Orthogonal complements	Orthogonal subspaces whose dimensions add to the ambient dimension.	Dot products zero and dimensions add correctly.

Common mistake. Do not say “orthogonal complement” from perpendicularity alone. Also check the dimension count.

3 Row Space and Nullspace

Let $A \in \mathbb{R}^{m \times n}$. Each row of A is a vector in $\mathbb{R}^n$. If $x \in N(A)$, then

$$Ax = 0.$$

This means every row of A has dot product zero with x . Therefore

$$C(A^T) \perp N(A).$$

Their dimensions add to the whole ambient space $\mathbb{R}^n$:

$$\dim C(A^T) + \dim N(A) = r + (n - r) = n.$$

So they are orthogonal complements:

$$N(A) = C(A^T)^\perp, \quad C(A^T) = N(A)^\perp.$$

4 Column Space and Left Nullspace

The left nullspace is

$$N(A^T) = \{y : A^T y = 0\}.$$

If $y \in N(A^T)$, then y is perpendicular to every column of A . Therefore

$$C(A) \perp N(A^T).$$

Their dimensions add to the whole ambient space $\mathbb{R}^m$:

$$\dim C(A) + \dim N(A^T) = r + (m - r) = m.$$

So they are orthogonal complements:

$$N(A^T) = C(A)^\perp, \quad C(A) = N(A^T)^\perp.$$

5 Four-Subspace Summary

Subspace	Lives in	Dimension	Orthogonal complement
Row space $C(A^T)$	$\mathbb{R}^n$	r	Nullspace $N(A)$.
Nullspace $N(A)$	$\mathbb{R}^n$	$n - r$	Row space $C(A^T)$.
Column space $C(A)$	$\mathbb{R}^m$	r	Left nullspace $N(A^T)$.
Left nullspace $N(A^T)$	$\mathbb{R}^m$	$m - r$	Column space $C(A)$.

Exam memory. The complement must live in the same ambient space: row with null in $\mathbb{R}^n$, column with left null in $\mathbb{R}^m$.

6 Small Example

Let

$$A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

The row space is

$$C(A^T) = \text{span}\{(1, 2, 0), (0, 0, 1)\}.$$

The nullspace is found from

$$x_1 + 2x_2 = 0, \quad x_3 = 0,$$

so

$$N(A) = \text{span}\{(-2, 1, 0)\}.$$

The dot products are zero:

$$(1, 2, 0) \cdot (-2, 1, 0) = 0, \quad (0, 0, 1) \cdot (-2, 1, 0) = 0.$$

This shows the row space direction and nullspace direction are perpendicular in $\mathbb{R}^3$.

7 How to Use This in Problems

If you know a vector is perpendicular to	Then place it in
Every row of A	$N(A)$.
Every vector in $N(A)$	$C(A^T)$.
Every column of A	$N(A^T)$.
Every vector in $N(A^T)$	$C(A)$.

Dimension check. Perpendicularity plus the correct dimension identifies the whole orthogonal complement.

8 Common Mistakes

Mistake	Correct exam habit
Putting row space in $\mathbb{R}^m$.	Row vectors have length n , so row space lives in $\mathbb{R}^n$.
Pairing column space with $N(A)$.	Column space pairs with $N(A^T)$.
Forgetting the left nullspace.	Solve $A^T y = 0$, not $Ax = 0$.
Calling any perpendicular subspaces complements.	Check that the dimensions add to n or m .
Copying the picture instead of using the dimension count.	Use the table and formulas to reconstruct the relationship.

9 Final Exam Checklist

Checklist.

1. Write the size: A is $m \times n$.
2. Identify the ambient space: $\mathbb{R}^n$ or $\mathbb{R}^m$.
3. Pair $C(A^T)$ with $N(A)$ inside $\mathbb{R}^n$.
4. Pair $C(A)$ with $N(A^T)$ inside $\mathbb{R}^m$.
5. Use $Ax = 0$ for the nullspace.
6. Use $A^T y = 0$ for the left nullspace.
7. Check dimension sums: $r + (n - r) = n$ and $r + (m - r) = m$.
8. State the orthogonal complement relationship directly.

4.2–4.3 Projections and Least Squares

Concise exam revision notes for projections, projection matrices, and least squares.

Based on handwritten MIT 18.06 revision notes.

1 Core Idea

Core idea. When $Ax = b$ has no solution, replace b by the closest vector p in the column space of A . The error $e = b - p$ must be perpendicular to that column space.

The picture is always

$$b = p + e, \quad p \in C(A), \quad e \in N(A^T).$$

Projection turns an unsolvable equation into a solvable normal equation.

2 Projection Onto a Line

For a nonzero vector a , the projection of b onto the line through a has the form

$$p = \hat{x}a.$$

The error is perpendicular to a :

$$a^T(b - \hat{x}a) = 0.$$

Solving gives

$$\hat{x} = \frac{a^T b}{a^T a}, \quad p = a \frac{a^T b}{a^T a}.$$

So the one-dimensional projection matrix is

$$P = \frac{aa^T}{a^T a}.$$

Meaning. $\hat{x}$ is the coordinate along the line; p is the closest point on the line.

3 Projection Onto a Subspace

Let the columns of A span the subspace. We look for

$$p = A\hat{x}.$$

The error $e = b - A\hat{x}$ must be perpendicular to every column of A :

$$A^T e = 0.$$

Therefore

$$A^T(b - A\hat{x}) = 0, \quad A^T A\hat{x} = A^T b.$$

These are the normal equations.

If the columns of A are independent, then $A^T A$ is invertible and

$$\hat{x} = (A^T A)^{-1} A^T b.$$

The projection is

$$p = A\hat{x} = A(A^T A)^{-1} A^T b.$$

So the projection matrix is

$$P = A(A^T A)^{-1} A^T.$$

4 Projection Matrix Facts

Fact	Formula	Meaning
Projection lands in $C(A)$	$p = Pb$	p is the closest vector in the column space.
Error is perpendicular	$A^T(b - p) = 0$	$e = b - p$ lies in $N(A^T)$.
Projecting twice changes nothing	$P^2 = P$	Once on the subspace, it stays there.
Orthogonal projection is symmetric	$P^T = P$	True for projection onto a column space.

Exam trigger. If the problem asks for closest vector, best fit, or least squares, expect $A^T A\hat{x} = A^T b$.

5 Orthogonal Columns Shortcut

If the columns $q_1, \dots, q_n$ are orthonormal, put them in Q . Then

$$Q^T Q = I.$$

Projection becomes much simpler:

$$p = Q Q^T b.$$

Equivalently,

$$p = (q_1^T b)q_1 + \dots + (q_n^T b)q_n.$$

Columns of A	Projection formula
Independent	$P = A(A^T A)^{-1} A^T$.
Orthogonal but not unit	Add projections along each column separately.
Orthonormal	$P = Q Q^T$.

Common mistake. Do not use $Q Q^T$ unless the columns are orthonormal. Orthogonal columns still need length factors.

6 Least Squares

Least squares solves the closest version of an inconsistent system:

$$Ax \approx b.$$

The best coefficient vector is $\hat{x}$, found from

$$A^T A \hat{x} = A^T b.$$

Then

$$p = A \hat{x}, \quad e = b - p.$$

For a best-fit line

$$y = C + Dt,$$

the matrix has rows

$$\begin{bmatrix} 1 & t_i \end{bmatrix}.$$

The unknown vector is

$$\hat{x} = \begin{bmatrix} C \\ D \end{bmatrix}.$$

The same normal equation gives the least-squares line.

7 P and $I - P$

The projection splits b into two perpendicular parts:

$$b = Pb + (I - P)b.$$

Here

$$Pb \in C(A), \quad (I - P)b \in N(A^T).$$

If P projects onto $C(A)$, then $I - P$ projects onto the left nullspace.

Exam memory. Projection onto a subspace plus projection onto its orthogonal complement gives the original vector.

8 Common Mistakes

Mistake	Correct exam habit
Solving $Ax = b$ when it is inconsistent.	Solve $A^T A \hat{x} = A^T b$.
Forgetting the error condition.	Check $A^T e = 0$.
Using $P = A^{-1}$ language.	Projection is not inversion. Use $P = A(A^T A)^{-1} A^T$.
Using $Q Q^T$ for non-orthonormal columns.	First check $Q^T Q = I$.
Assuming $P_1 + P_2 = I$ for any two projections.	This works for orthogonal complementary subspaces.

9 Final Exam Checklist

Checklist.

1. Identify the subspace: usually $C(A)$ or a line through a .
2. Write $p = A\hat{x}$ or $p = \hat{x}a$.
3. Write the error $e = b - p$.
4. Impose perpendicularity: $A^T e = 0$ or $a^T e = 0$.
5. Solve $A^T A\hat{x} = A^T b$.
6. Compute $p = A\hat{x}$.
7. Compute the error $e = b - p$.
8. Use $P = A(A^T A)^{-1}A^T$ only when the columns of A are independent.
9. Use $P = QQ^T$ only when the columns are orthonormal.
10. State where p and e live: $p \in C(A)$, $e \in N(A^T)$.

4.4, 5.1–5.2 Orthonormal Bases, QR, and Determinants

Concise exam revision notes for orthonormal columns, QR factorization, and determinant rules.

Based on handwritten MIT 18.06 revision notes.

1 Core Idea

Core idea. Orthonormal columns make projection and least squares simple. Determinants measure whether a square matrix is singular and how elimination changes volume.

This note has two connected parts:

$$Q^T Q = I \quad \text{and} \quad \det A.$$

The first simplifies geometry; the second summarizes invertibility for square matrices.

2 Orthogonal vs Orthonormal

Word	Condition	Meaning
Orthogonal	$q_i^T q_j = 0$ for $i \neq j$.	Different directions are perpendicular.
Orthonormal	$q_i^T q_j = 0$ and $q_i^T q_i = 1$.	Perpendicular and unit length.
Matrix Q with orthonormal columns	$Q^T Q = I$.	Dot products are already built in.

If Q is square and orthonormal, then

$$Q^{-1} = Q^T.$$

For rectangular Q , $Q^T Q = I$, but $Q Q^T$ is a projection matrix.

3 Gram–Schmidt

Gram–Schmidt turns independent vectors into orthogonal vectors, then normalizes them.

For two vectors a, b , first keep

$$A = a.$$

Remove from b the part in the a -direction:

$$B = b - \frac{a^T b}{a^T a} a.$$

Then normalize:

$$q_1 = \frac{A}{\|A\|}, \quad q_2 = \frac{B}{\|B\|}.$$

For a third vector c , subtract the projections onto the previous orthogonal directions:

$$C = c - \frac{A^T c}{A^T A} A - \frac{B^T c}{B^T B} B.$$

Exam trigger. If the problem says “make an orthonormal basis”, use Gram–Schmidt, then divide by lengths.

4 QR Factorization

For independent columns,

$$A = QR,$$

where Q has orthonormal columns and R is upper triangular.

The entries of R record dot products with the q ’s:

$$R = Q^T A.$$

For least squares,

$$A^T A \hat{x} = A^T b$$

becomes

$$R \hat{x} = Q^T b.$$

This is easier because R is triangular.

5 Projection with Orthonormal Columns

If Q has orthonormal columns, projection onto $C(Q)$ is

$$p = QQ^T b.$$

In vector form,

$$p = (q_1^T b)q_1 + (q_2^T b)q_2 + \cdots + (q_n^T b)q_n.$$

Column type	Projection habit
Orthonormal q_i	Use $q_i^T b$ directly.
Orthogonal but not unit	Divide by $a_i^T a_i$.
General independent columns	Use $A(A^T A)^{-1} A^T$.

6 Orthogonal Matrices

For square orthogonal Q ,

$$Q^T Q = I, \quad Q^{-1} = Q^T.$$

Orthogonal matrices preserve lengths and dot products:

$$\|Qx\| = \|x\|, \quad (Qx)^T (Qy) = x^T y.$$

Rotations and reflections are the main exam examples.

7 Determinant Rules

Operation or fact	Effect on determinant
Exchange two rows	Changes the sign.
Multiply one row by c	Multiplies determinant by c .
Add a multiple of one row to another	No change.
Triangular matrix	Product of diagonal entries.
Singular matrix	Determinant is 0.
Product	$\det(AB) = \det A \det B$.
Transpose	$\det(A^T) = \det A$.

Exam memory. A zero determinant means no inverse. A nonzero determinant means the square matrix is invertible.

8 Pivots and Cofactors

If elimination uses no row exchanges, then

$$\det A = (\text{product of pivots}).$$

For leading submatrices A_k , pivots can be checked by determinant ratios:

$$d_k = \frac{\det A_k}{\det A_{k-1}}.$$

Cofactor expansion is useful when a row or column has many zeros:

$$\det A = \sum_j (-1)^{i+j} a_{ij} \det M_{ij}.$$

Use it selectively; elimination is usually faster for larger matrices.

9 Common Mistakes

Mistake	Correct exam habit
Calling orthogonal columns orthonormal.	Check both perpendicularity and unit length.
Using $Q^{-1} = Q^T$ for rectangular Q .	Only square orthogonal matrices have inverse Q^T .
Forgetting to normalize after Gram–Schmidt.	Orthogonal first, unit length second.
Using QQ^T for any basis.	Use it only for orthonormal columns.
Changing a row and forgetting the determinant effect.	Track swaps, scaling, and row replacement separately.
Expanding cofactors when elimination is simpler.	Use zeros or structure to choose the method.

10 Final Exam Checklist

Checklist.

1. Check whether vectors are orthogonal by dot products.
2. Check whether vectors are orthonormal by dot products and lengths.
3. Use Gram–Schmidt to build orthogonal vectors.
4. Normalize each vector to get q_i .
5. Form Q from the q_i 's and compute $R = Q^T A$.
6. For least squares with $A = QR$, solve $R\hat{x} = Q^T b$.
7. For determinants, choose elimination unless cofactor expansion has many zeros.
8. Track determinant changes from row operations.
9. Use $\det A = 0$ to identify singular square matrices.

5.3 Inverses, Cramer's Rule, Cross Products, and Volume

Concise exam revision notes for adjugates, determinant geometry, and orientation.

Based on handwritten MIT 18.06 revision notes.

1 Core Idea

Core idea. Determinants connect algebra and geometry: they decide invertibility, give formulas for inverse entries, and measure signed area or volume.

For an invertible square matrix,

$$A^{-1} = \frac{1}{\det A} C^T,$$

where C is the cofactor matrix. Equivalently,

$$(\det A)I = AC^T.$$

2 Cramer's Rule

For

$$Ax = b,$$

each variable is a determinant ratio:

$$x_i = \frac{\det A_i(b)}{\det A},$$

where $A_i(b)$ is A with column i replaced by b .

Use Cramer's rule when	Avoid it when
The matrix is small or symbolic. You need a formula for one variable.	The matrix is large and elimination is easier. You need all variables numerically.

Common mistake. Cramer's rule requires $\det A \neq 0$. If $\det A = 0$, the determinant ratios do not apply.

3 Area from Determinants

For two vectors in the plane,

$$u = \begin{bmatrix} a \\ b \end{bmatrix}, \quad v = \begin{bmatrix} c \\ d \end{bmatrix},$$

the signed area of the parallelogram is

$$\det \begin{bmatrix} a & c \\ b & d \end{bmatrix} = ad - bc.$$

The absolute value gives the geometric area:

$$\text{area} = |ad - bc|.$$

4 Cross Product

In $\mathbb{R}^3$, the cross product is perpendicular to both input vectors:

$$u \times v \perp u, \quad u \times v \perp v.$$

Using unit vectors,

$$u \times v = \det \begin{bmatrix} i & j & k \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{bmatrix}.$$

Its length is the parallelogram area:

$$\|u \times v\| = \|u\| \|v\| \sin \theta.$$

Exam use. Use cross product when a problem asks for a normal vector to a plane in $\mathbb{R}^3$.

5 Orientation

The determinant sign records orientation:

$$\det A > 0$$

preserves orientation, while

$$\det A < 0$$

reverses orientation.

The scalar triple product gives signed volume:

$$\det \begin{bmatrix} u_1 & v_1 & w_1 \\ u_2 & v_2 & w_2 \\ u_3 & v_3 & w_3 \end{bmatrix} = (u \times v) \cdot w.$$

If this determinant is zero, the three vectors lie in the same plane.

6 Common Mistakes

Mistake	Correct exam habit
Using Cramer's rule when $\det A = 0$.	First check invertibility.
Forgetting the transpose in the inverse formula.	Use $A^{-1} = C^T / \det A$.
Dropping the sign of area too early.	Use signed area for orientation; absolute value for size.
Writing $u \times v = v \times u$.	Cross product changes sign: $v \times u = -(u \times v)$.
Using cross product outside $\mathbb{R}^3$.	The standard cross product is a 3D tool.

7 Final Exam Checklist

Checklist.

1. Check whether $\det A \neq 0$ before using inverse formulas.
2. Build the cofactor matrix C .
3. Use $A^{-1} = C^T / \det A$.
4. For Cramer's rule, replace one column at a time by b .
5. Use determinant sign for orientation.
6. Use absolute determinant for area or volume size.
7. Use $u \times v$ for a normal vector in $\mathbb{R}^3$.
8. Check $u \cdot (u \times v) = 0$ and $v \cdot (u \times v) = 0$.

6.1–6.2 Eigenvalues, Eigenvectors, and Diagonalization

Concise exam revision notes for eigenvectors, characteristic equations, powers, and stability.

Based on handwritten MIT 18.06 revision notes.

1 Core Idea

Core idea. Most vectors change direction when multiplied by A . Eigenvectors keep their direction; only their length and sign are scaled.

The defining equation is

$$Ax = \lambda x, \quad x \neq 0.$$

Equivalently,

$$(A - \lambda I)x = 0.$$

So $A - \lambda I$ must be singular:

$$\det(A - \lambda I) = 0.$$

2 How to Find Eigenvalues and Eigenvectors

1. Solve $\det(A - \lambda I) = 0$ to find eigenvalues.
2. For each eigenvalue, solve $(A - \lambda I)x = 0$.
3. Use any nonzero vector in that nullspace as an eigenvector.

Exam trigger. Eigenvectors come from a nullspace. If the nullspace has more than one direction, choose a convenient basis.

3 Fast Checks

Check	Formula	Meaning
Trace	$\text{trace}(A) = \lambda_1 + \cdots + \lambda_n$	Sum of eigenvalues.
Determinant	$\det A = \lambda_1 \lambda_2 \cdots \lambda_n$	Product of eigenvalues.
Invertibility	0 is not an eigenvalue	Equivalent to $\det A \neq 0$.

These checks catch arithmetic mistakes after solving the characteristic equation.

4 Special Matrices

Matrix type	Typical eigenvalues	Exam memory
Projection P	0 and 1	$P^2 = P$.
Reflection R	1 and -1	Directions along and across the mirror.
Rotation in the plane	Usually complex	Real rotations may have no real eigenvectors.
Triangular matrix	Diagonal entries	Read eigenvalues from the diagonal.

Common mistake. Row reduction changes eigenvalues in general. Do not row-reduce A and read eigenvalues from the new matrix.

5 Diagonalization

If A has n independent eigenvectors, put them into

$$S = \begin{bmatrix} x_1 & x_2 & \cdots & x_n \end{bmatrix}.$$

Then

$$AS = S\Lambda, \quad A = S\Lambda S^{-1}.$$

Here

$$\Lambda = \begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{bmatrix}.$$

Case	Diagonalization result
n distinct eigenvalues	Automatically enough independent eigenvectors.
Repeated eigenvalue	May or may not have enough eigenvectors.
Not enough eigenvectors	Not diagonalizable.

Exam memory. Repeated eigenvalues are not the problem by themselves. The issue is whether there are enough independent eigenvectors.

6 Powers and Difference Equations

Diagonalization makes powers easy:

$$A^k = S\Lambda^k S^{-1}.$$

For a difference equation

$$u_{k+1} = Au_k,$$

the solution is

$$u_k = A^k u_0 = S\Lambda^k S^{-1} u_0.$$

If

$$u_0 = c_1 x_1 + \cdots + c_n x_n,$$

then

$$u_k = c_1 \lambda_1^k x_1 + \cdots + c_n \lambda_n^k x_n.$$

7 Stability

Eigenvalue behavior	Effect on A^k
All $ \lambda_i < 1$	Decays to zero.
Some $ \lambda_i > 1$	Blows up in that direction.
$ \lambda_i = 1$	Borderline; behavior depends on the matrix.
One $\lambda = 1$, rest $ \lambda < 1$	Possible steady state.

8 Common Mistakes

Mistake	Correct exam habit
Solving $Ax = 0$ for every eigenvector.	Solve $(A - \lambda I)x = 0$.
Forgetting $x \neq 0$.	Eigenvectors are never the zero vector.
Assuming $\lambda = 0$ means diagonalizable fails.	It means A is singular; diagonalizability is separate.
Assuming repeated eigenvalues always fail.	Count independent eigenvectors.
Row-reducing A before finding eigenvalues.	Use $\det(A - \lambda I) = 0$.
Confusing invertibility and diagonalizability.	Invertible means no zero eigenvalue; diagonalizable means enough eigenvectors.

9 Final Exam Checklist

Checklist.

1. Compute $\det(A - \lambda I)$.
2. Solve for all eigenvalues λ .
3. Check trace and determinant against the eigenvalues.
4. For each λ , solve $(A - \lambda I)x = 0$.
5. Count independent eigenvectors.
6. Diagonalize only if there are n independent eigenvectors.
7. Form S from eigenvectors and Λ from eigenvalues.
8. Use $A^k = S\Lambda^k S^{-1}$ for powers.
9. Use $|\lambda|$ to decide growth or decay in difference equations.

Eigenvalue Quick Reference

Concise revision notes for eigenvalue equations, checks, and basic matrix types.

Based on handwritten MIT 18.06 revision notes.

1 Core Idea

Core idea. Eigenvectors are directions that survive multiplication by A . The matrix only stretches, shrinks, reverses, or rotates through a scalar factor.

The defining equation is

$$Ax = \lambda x, \quad x \neq 0.$$

Move all terms to one side:

$$(A - \lambda I)x = 0.$$

For a nonzero solution, the matrix $A - \lambda I$ must be singular:

$$\det(A - \lambda I) = 0.$$

2 Characteristic Equation

For a 2×2 matrix, the characteristic equation can be written as

$$\lambda^2 - \text{trace}(A)\lambda + \det A = 0.$$

This is often the fastest way to check arithmetic.

Quantity	Eigenvalue check
Trace	Sum of eigenvalues.
Determinant	Product of eigenvalues.
Zero eigenvalue	Matrix is singular.

3 Basic Examples

Matrix type	Eigenvalue memory
Projection P with $P^2 = P$	Eigenvalues are 0 or 1.
Permutation swapping two coordinates	Eigenvalues often include 1 and -1 .
Plane rotation by 90°	Eigenvalues are complex: i and $-i$.

4 Growth and Stability

For repeated multiplication A^k :

$$|\lambda| < 1$$

means decay in that eigenvector direction.

$$|\lambda| > 1$$

means growth in that direction.

$$\lambda = 1$$

means that direction stays fixed.

5 Common Mistakes

Mistake	Correct exam habit
Allowing $x = 0$ as an eigenvector.	Eigenvectors must be nonzero.
Solving $Ax = 0$ for every eigenvector.	Solve $(A - \lambda I)x = 0$.
Forgetting trace and determinant checks.	Use them after solving for λ .
Assuming all rotations have real eigenvectors.	Plane rotations can have complex eigenvalues.

6 Final Exam Checklist

Checklist.

1. Write $Ax = \lambda x$.
2. Form $A - \lambda I$.
3. Solve $\det(A - \lambda I) = 0$.
4. Check trace and determinant.
5. For each λ , solve $(A - \lambda I)x = 0$.
6. Use $|\lambda|$ to judge growth or decay for powers.

6.3–6.5 Matrix Exponentials, Symmetric Matrices, and Positive Definiteness

Concise exam revision notes for continuous systems, symmetric eigenvectors, and positive definite tests.

Based on handwritten MIT 18.06 revision notes.

1 Core Idea

Core idea. Eigenvectors decouple matrix problems. For symmetric matrices, the decoupling is especially clean because the eigenvectors can be chosen orthonormal.

The same eigenvalue idea controls:

$$\frac{du}{dt} = Au, \quad S = Q\Lambda Q^T, \quad x^T Sx.$$

2 Differential Equations

For the system

$$\frac{du}{dt} = Au,$$

if A has independent eigenvectors $x_1, \dots, x_n$, then the solution has the form

$$u(t) = c_1 e^{\lambda_1 t} x_1 + \dots + c_n e^{\lambda_n t} x_n.$$

The constants come from the initial condition:

$$Sc = u(0), \quad c = S^{-1}u(0).$$

In matrix form,

$$u(t) = e^{At}u(0), \quad e^{At} = Se^{\Lambda t}S^{-1}.$$

Here

$$e^{\Lambda t} = \begin{bmatrix} e^{\lambda_1 t} & 0 & \dots & 0 \\ 0 & e^{\lambda_2 t} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & e^{\lambda_n t} \end{bmatrix}.$$

3 Continuous Stability

Eigenvalues of A	Behavior of e^{At}
All $\operatorname{Re} \lambda < 0$	Decays to zero.
Some $\operatorname{Re} \lambda > 0$	Blows up in that direction.
One $\lambda = 0$, rest $\operatorname{Re} \lambda < 0$	Possible steady state.
Complex $\lambda = a + ib$	a controls growth/decay; b controls oscillation.

Exam memory. For powers use $|\lambda|$. For differential equations use $\operatorname{Re} \lambda$.

4 Defective Matrices

A repeated eigenvalue may not give enough eigenvectors. Then S is not invertible, so

$$A = SAS^{-1}$$

is impossible.

For a defective repeated eigenvalue, a generalized eigenvector can replace a missing eigenvector:

$$(A - \lambda I)w = x.$$

This creates terms involving $te^{\lambda t}$ in differential equations.

Scope. For exam revision, first try ordinary eigenvectors. Use generalized eigenvectors only when the matrix is defective.

5 Symmetric Matrices

For a real symmetric matrix

$$S = S^T.$$

The key facts are:

- all eigenvalues are real;
- eigenvectors for different eigenvalues are orthogonal;
- the eigenvectors can be chosen orthonormal.

Therefore a symmetric matrix factors as

$$S = Q\Lambda Q^T.$$

Equivalently,

$$S = \lambda_1 q_1 q_1^T + \cdots + \lambda_n q_n q_n^T.$$

Exam trigger. If the matrix is symmetric, look for orthogonal eigenvectors and use $Q^T = Q^{-1}$.

6 Positive Definite Matrices

For a symmetric matrix S , positive definite means

$$x^T S x > 0 \quad \text{for every } x \neq 0.$$

Equivalent tests:

Test	Positive definite condition
Eigenvalues	All $\lambda_i > 0$.
Pivots	All pivots are positive.
Leading determinants	All leading principal determinants are positive.
Quadratic form	$x^T S x > 0$ for all $x \neq 0$.
Factorization	$S = R^T R$ with independent columns in R .

For 2×2

$$S = \begin{bmatrix} a & b \\ b & c \end{bmatrix},$$

the quick test is

$$a > 0, \quad ac - b^2 > 0.$$

7 Positive Semidefinite

Positive semidefinite means

$$x^T S x \geq 0 \quad \text{for every } x.$$

Then eigenvalues are nonnegative:

$$\lambda_i \geq 0.$$

It may be singular:

$$\det S = 0$$

is allowed.

Common mistake. Positive semidefinite is not the same as positive definite. Zero eigenvalues are allowed only in the semidefinite case.

8 Similarity and Transpose

Similar matrices have the form

$$B = M^{-1} A M.$$

They have the same eigenvalues because they represent the same linear map in different coordinates.

A matrix and its transpose have the same eigenvalues:

$$\det(A^T - \lambda I) = \det(A - \lambda I).$$

But their eigenvectors do not have to be the same.

9 Quadratic Forms and Axes

A symmetric matrix rotates quadratic forms into principal axes:

$$x^T S x = (Q^T x)^T \Lambda (Q^T x).$$

For an ellipse, the eigenvectors give the axis directions and the eigenvalues control the axis lengths.

10 Common Mistakes

Mistake	Correct exam habit
Using $ \lambda $ for differential equation stability.	Use $\operatorname{Re} \lambda$.
Assuming repeated eigenvalues give enough eigenvectors.	Check the eigenspace dimension.
Using $Q \Lambda Q^T$ for a non-symmetric matrix.	Reserve that clean form for symmetric matrices.
Testing positive definiteness without symmetry.	The standard tests here assume a symmetric matrix.
Confusing semidefinite with definite.	Positive definite needs strict positivity and no zero eigenvalues.
Assuming A and A^T have the same eigenvectors.	They have the same eigenvalues, not necessarily the same eigenvectors.

11 Final Exam Checklist

Checklist.

1. For $u' = Au$, find eigenvalues and eigenvectors of A .
2. Write $u(t) = \sum c_i e^{\lambda_i t} x_i$ when A has enough eigenvectors.
3. Use $Sc = u(0)$ to find constants.
4. Use $\operatorname{Re} \lambda$ to decide continuous stability.
5. Check symmetry before using $S = Q \Lambda Q^T$.
6. For a symmetric matrix, choose eigenvectors orthonormal.
7. Test positive definiteness by eigenvalues, pivots, leading determinants, or $x^T S x$.
8. Treat zero eigenvalues as semidefinite, not definite.
9. Use similarity to preserve eigenvalues, not necessarily eigenvectors.

Eigenvalue Patterns for Common Matrix Types

Concise exam revision table for recognizing eigenvalue behavior quickly.

Based on handwritten MIT 18.06 revision notes.

1 Core Idea

Core idea. Special matrix structure often predicts eigenvalue behavior before calculation. Use structure first, then verify by computation.

This note is an original compact reference based on standard 18.06 facts.

2 Common Patterns

Matrix type	Eigenvalue pattern	Exam use
Symmetric $S = S^T$	Real eigenvalues.	Orthogonal eigenvectors; use $S = Q\Lambda Q^T$.
Orthogonal $Q^T Q = I$	$ \lambda = 1$.	Lengths are preserved.
Skew-symmetric $A^T = -A$	Pure imaginary or zero eigenvalues.	Diagonal entries are zero.
Projection $P^2 = P$	$\lambda = 0$ or 1 .	Range stays, nullspace collapses.
Positive definite S	All $\lambda > 0$.	Use for quadratic forms and pivots.
Rank-one uv^T	At most one nonzero eigenvalue.	Most directions collapse.
Triangular	Diagonal entries.	Read eigenvalues directly.
Similar matrices	Same eigenvalues.	Change coordinates without changing spectrum.

3 Shift, Inverse, and Powers

If $Ax = \lambda x$, then:

$$(A + cI)x = (\lambda + c)x.$$

So shifting by cI shifts eigenvalues by c .

If A is invertible, then:

$$A^{-1}x = \frac{1}{\lambda}x.$$

So inverse eigenvalues are reciprocals.

For powers:

$$A^k x = \lambda^k x.$$

This is why eigenvalues control long-term behavior.

4 Stability Memory

Problem type	Stability test
Repeated powers A^k	Use $ \lambda $.
Differential equation $u' = Au$	Use $\operatorname{Re} \lambda$.
Projection	Stable because eigenvalues are 0, 1.
Orthogonal iteration	Lengths do not grow because $ \lambda = 1$.

5 Common Mistakes

Mistake	Correct exam habit
Using $S = Q\Lambda Q^T$ without symmetry.	First check $S = S^T$.
Assuming all real matrices have real eigenvalues.	Rotations can give complex eigenvalues.
Forgetting zero eigenvalues in projections.	Projection kills directions in the nullspace.
Using $ \lambda $ for $u' = Au$.	Use $\operatorname{Re} \lambda$ for continuous systems.
Assuming same eigenvalues means same eigenvectors.	Similar matrices preserve eigenvalues, not necessarily vector coordinates.

6 Final Exam Checklist

Checklist.

1. Identify special structure before calculating.
2. Check symmetry, orthogonality, projection, or triangular form.
3. Use the structure to predict eigenvalue behavior.
4. Verify with trace, determinant, or $\det(A - \lambda I)$.
5. Use shifts, inverses, and powers to transform known eigenvalues.
6. Use $|\lambda|$ for powers and $\operatorname{Re} \lambda$ for differential equations.

Jordan Blocks and Defective Matrices

Exam notes for repeated eigenvalues, missing eigenvectors, and Jordan form.

Based on handwritten MIT 18.06 revision notes.

1 Core Idea

Core idea. Repeated eigenvalues do not guarantee enough eigenvectors. When a matrix is missing eigenvectors, Jordan blocks replace diagonal form.

Diagonalization is the best case:

$$A = S\Lambda S^{-1}.$$

This works only when A has a full basis of independent eigenvectors.

2 Diagonalizable vs Defective

Case	Meaning	Exam signal
Diagonalizable	Enough independent eigenvectors to form S .	Each eigenspace supplies the needed number of vectors.
Defective	Not enough independent eigenvectors.	Some repeated eigenvalue has too small a nullspace.
Jordan form	Replacement for diagonal form.	Ones appear just above the diagonal inside a block.

3 Algebraic and Geometric Multiplicity

For an eigenvalue λ :

$$\begin{aligned}m_a(\lambda) &= \text{multiplicity of } \lambda \text{ in } \det(A - \lambda I), \\m_g(\lambda) &= \dim N(A - \lambda I).\end{aligned}$$

The exam test is:

$$A \text{ is diagonalizable} \iff m_g(\lambda) = m_a(\lambda) \text{ for every } \lambda.$$

If $m_g(\lambda) < m_a(\lambda)$, the matrix is defective for that eigenvalue.

4 Jordan Blocks

A Jordan block for eigenvalue λ has λ on the diagonal and ones on the superdiagonal:

$$J_\lambda = \begin{bmatrix} \lambda & 1 & 0 \\ 0 & \lambda & 1 \\ 0 & 0 & \lambda \end{bmatrix}.$$

The diagonal part gives the eigenvalue. The superdiagonal ones record the missing eigenvector directions.

5 Jordan Form

Instead of diagonalizing, a defective matrix may be written as

$$A = MJM^{-1},$$

where J is block diagonal with Jordan blocks.

A diagonal matrix is the special case where every Jordan block has size 1.

6 Generalized Eigenvectors

If x is an eigenvector, then

$$(A - \lambda I)x = 0.$$

A generalized eigenvector w may satisfy

$$(A - \lambda I)w = x.$$

Together, w and x form a chain. This chain supplies the columns of M for the Jordan block.

7 Exam Method

1. Find eigenvalues from $\det(A - \lambda I) = 0$.
2. For each repeated eigenvalue, compute $N(A - \lambda I)$.
3. Compare $m_g(\lambda)$ with $m_a(\lambda)$.
4. If all match, diagonalize.
5. If one fails, use Jordan blocks for that eigenvalue.

8 Common Mistakes

Mistake	Correct exam habit
Assuming repeated eigenvalues mean defective.	Check the dimension of the eigenspace.
Assuming distinct eigenvalues are required.	Repeated eigenvalues can still be diagonalizable.
Counting algebraic multiplicity as eigenvectors.	Count independent vectors in $N(A - \lambda I)$.
Forgetting the superdiagonal ones in a Jordan block.	A nontrivial block has ones above the diagonal.
Treating $A = MJM^{-1}$ as diagonalization.	Jordan form is weaker than diagonal form.

9 Final Exam Checklist

Checklist.

1. Compute the characteristic polynomial.
2. List algebraic multiplicities.
3. Compute eigenspace dimensions.
4. Compare $m_a(\lambda)$ and $m_g(\lambda)$.
5. Use diagonal form if enough eigenvectors exist.
6. Use Jordan blocks when eigenvectors are missing.
7. Check that the total block sizes add to n .

7.2 Singular Value Decomposition

Exam notes for $A = U\Sigma V^T$, singular values, and the four subspaces.

Based on handwritten MIT 18.06 revision notes.

1 Core Idea

Core idea. The SVD writes any matrix as orthogonal input directions, nonnegative stretching, and orthogonal output directions.

For an $m \times n$ matrix,

$$A = U\Sigma V^T.$$

Here V gives orthonormal directions in $\mathbb{R}^n$, U gives orthonormal directions in $\mathbb{R}^m$, and Σ contains the singular values.

2 What the SVD Says

$$Av_i = \sigma_i u_i.$$

This means: the right singular vector v_i is sent to the left singular vector u_i , scaled by $\sigma_i \geq 0$.

Object	Lives in	Meaning
v_i	$\mathbb{R}^n$	Orthonormal input direction.
u_i	$\mathbb{R}^m$	Orthonormal output direction.
σ_i	Nonnegative number	Stretch factor from v_i to u_i .
Σ	$m \times n$ diagonal shape	Stores singular values in decreasing order.

3 How to Compute It

The right singular vectors come from the symmetric matrix $A^T A$:

$$A^T A v_i = \sigma_i^2 v_i.$$

So the singular values are

$$\sigma_i = \sqrt{\lambda_i(A^T A)}.$$

For each $\sigma_i > 0$, compute the matching left singular vector by

$$u_i = \frac{A v_i}{\sigma_i}.$$

If U or V is not complete, add orthonormal vectors to finish the basis.

4 Geometry

The SVD is the clean geometric order:

$$x \xrightarrow{V^T} \text{rotate/reflection} \xrightarrow{\Sigma} \text{stretch} \xrightarrow{U} \text{rotate/reflection}.$$

The singular values show the principal stretches of the matrix.

5 Rank and the Four Subspaces

If $\sigma_1, \dots, \sigma_r > 0$ and the rest are zero, then $r = \text{rank}(A)$.

Subspace	Basis from the SVD
Row space $C(A^T)$	$v_1, \dots, v_r$.
Nullspace $N(A)$	Remaining v_i with $\sigma_i = 0$.
Column space $C(A)$	$u_1, \dots, u_r$.
Left nullspace $N(A^T)$	Remaining u_i .

6 SVD vs Eigenvalue Decomposition

Feature	Eigenvalue decomposition	SVD
Basic form	$A = S\Lambda S^{-1}$	$A = U\Sigma V^T$
Works for	Diagonalizable square matrices	Every matrix
Numbers	Eigenvalues may be negative or complex	Singular values are nonnegative real numbers
Directions	Same space in and out	Input directions v_i , output directions u_i
Best use	Dynamics and powers	Geometry, rank, least squares, approximation

7 Symmetric Case

For a symmetric positive definite matrix,

$$A = Q\Lambda Q^T$$

already has orthonormal eigenvectors and positive eigenvalues. In this special case, the SVD agrees with the eigenvalue decomposition.

For a symmetric matrix with negative eigenvalues, singular values use absolute sizes, not signs.

8 Common Mistakes

Mistake	Correct exam habit
Treating singular values as eigenvalues of A .	Singular values come from $A^T A$.
Allowing negative singular values.	Always take $\sigma_i \geq 0$.
Using $u_i = Av_i/\sigma_i$ when $\sigma_i = 0$.	Only divide by positive singular values.
Forgetting rectangular dimensions.	Σ has the same shape as A .
Mixing up U and V .	V belongs to the input side; U belongs to the output side.

9 Final Exam Checklist

Checklist.

1. Compute $A^T A$.
2. Find eigenvalues and orthonormal eigenvectors of $A^T A$.
3. Set $\sigma_i = \sqrt{\lambda_i}$.
4. Put the v_i 's into V .
5. Compute $u_i = Av_i/\sigma_i$ for positive σ_i .
6. Complete orthonormal bases when needed.
7. Write $A = U\Sigma V^T$ with matching dimensions.
8. Use zero singular values to identify nullspaces.